

Conditional Probability.

eg:

	Married	Unmarried	Total
Male	75	25	(100) → Total no. of males
Female	90	10	(100) → Total no. of females
Total	(165) ↓ Total no. of married person	(35) ↓ Total no. of unmarried.	(200) ↓ Total no. of person ↓ A person is selected ↓ Random Experiment

Total no. of case, $n = 200$

Individual Events:

A = Married

B = Unmarried

C = Male

D = Female

Then, $P(A) = \frac{165}{200}$

$P(C) = \frac{100}{200}$

$P(B) = \frac{35}{200}$

$P(D) = \frac{100}{200}$

probability of individual cases

Joint - events: / Compound Events

$A \cap C$ = Person is married and male

$A \cap D$ = Married and female

$B \cap C$ = Unmarried and male

$B \cap D$ = Unmarried and female

Then, $P(A \cap C) = \frac{75}{200}$

$$P(A \cap D) = \frac{90}{200}$$

$$P(B \cap C) = \frac{25}{200}$$

$$P(B \cap D) = \frac{10}{200}$$

Now,

if a person is selected and he is
male $\rightarrow$ conditional prob
additional info.

$P(A|C)$ { A given C }

and sample space decreases

$\hookrightarrow$ Prob (married / male)

$\hookrightarrow$ problem of being married
given he is a male.

$$P(A|C) = \frac{75}{200} \text{ total cases}$$

$$P(B|C) = \frac{25}{100}$$

we can see the condn is about male. $P(A|D)$ or

The Conditional probability of A given that B has already occurred is denoted by, $P(A|B)$ and is given as;

$$P(A|B) = \frac{P(A \cap B)}{P(B)} ; \text{ provided that } P(B) \neq 0$$

← joint prob.
← marginal prob.

Similarly, the conditional probability of B given that A has already occurred is denoted by $P(B|A)$ and is given as;

$$P(B|A) = \frac{P(A \cap B)}{P(A)} ; P(A) \neq 0$$

Q) If the probability that a communication system will have high fidelity is 0.81 and the probability that it will have high fidelity and high selectivity is 0.18. what is the probability that a system with high fidelity will also have high selectivity.

Sol. Define the events;
 A = System have high fidelity
 B = System have high selectivity

Then

$$P(A) = 0.81$$

$$P(A \cap B) = 0.18$$

Now, probability that a system with high fidelity will also have high selectivity.

It is asking for high selectivity
if it has high fidelity.

→ Probability that system has high selectivity given that it has high fidelity.

$$\therefore, P(B/A) = \frac{P(A \cap B)}{P(A)}$$

→ Bayes' Theorem (Bayes' Rule)

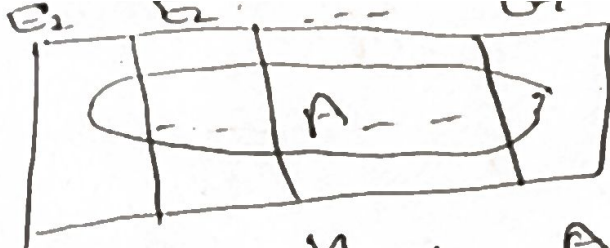
- Developed by Thomas Bayes.
- Extension of Conditional probability
- Inverse probability

Bayes' Theorem is a method of calculating updated or revised or inverse probability (posterior probability) of the previous or prior probability on the basis of additional information (conditional probability).

Statement:

Let $E_1, E_2, \dots, E_n$ be n mutually exclusive events defined on sample space (S) with $P(E_i) \neq 0, i = 1, 2, \dots, n$. For any arbitrary event A which is the subset of sample space (S) such that

$$P(A) > 0$$



Then, prob. of occurrence of E_i ($i=1, 2, \dots, n$) given that A has already occurred is denoted by $P(E_i/A)$ and is given as:

$$P(E_i/A) = \frac{P(E_i) \cdot P(A/E_i)}{\sum_{i=1}^n P(E_i) \cdot P(A/E_i)}$$

where, $P(E_i)$ = Prior probabilities
(Information before Experiment)

$P(A/E_i)$ = Conditional probability
(Additional info.)

$P(E_i/A)$ = Posterior prob.

$\Downarrow$
Inverse prob.
(Updated prob.)
of E_i
on condition
A.

Proof:

From, the definition of conditional probability;

$$P(E_i/A) = \frac{P(E_i \cap A)}{P(A)}$$

$$\left\{ \begin{aligned} \text{Also: } P(A \cap B) &= P(A) \cdot P(B/A) \\ &= P(B) \cdot P(A/B) \end{aligned} \right\}$$

$$= \frac{P(E_i) \cdot P(A|E_i)}{P(A)} \rightarrow \textcircled{9}$$

Here, $E_1, E_2, \dots, E_n$ are mutually exclusive events defines on sample space (S). Then, sample space (S) can be written as;

$$S = E_1 \cup E_2 \cup \dots \cup E_n$$

Also, A is the subset of same space (S). Then,

$$A = A \cap S$$

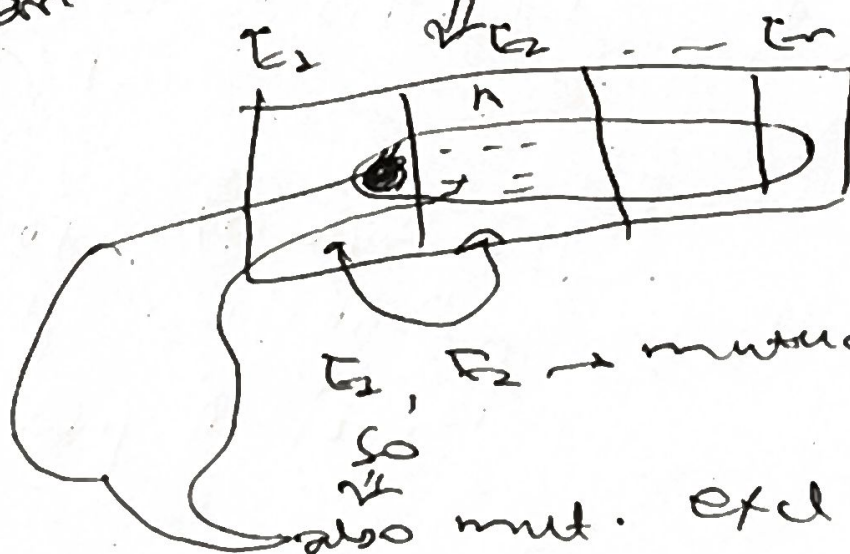
$$= A \cap [E_1 \cup E_2 \cup \dots \cup E_n]$$

$$= (A \cap E_1) \cup (A \cap E_2) \cup \dots$$

$$\cup (A \cap E_n) \quad [\because \text{distributive law}] \rightarrow \textcircled{10}$$

Again,

$E_1, E_2, \dots, E_n$ are mutually exclusive events. Then, $(A \cap E_1), (A \cap E_2), \dots$ are also mutually exclusive.



Now, probability on both sides of eqn.

(ii), we get,

$$P(A) = P[(A \cap E_1) \cup (A \cap E_2) \cup \dots \cup (A \cap E_n)]$$

$$= P(A \cap E_1) + P(A \cap E_2) + \dots + P(A \cap E_n)$$

$$= \sum_{i=1}^n P(A \cap E_i)$$

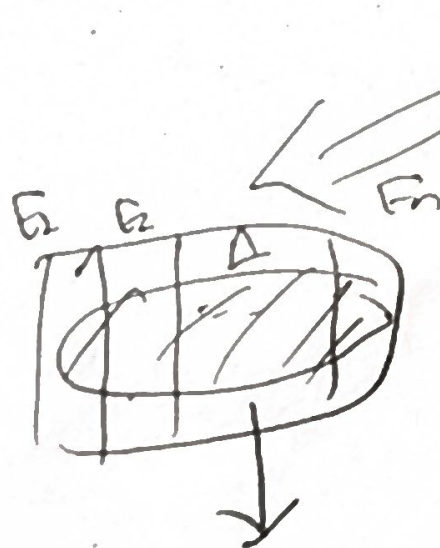
$\therefore$ All are mutually exclusion

$$\text{then } P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

$$\Rightarrow P(A) = \sum_{i=1}^n P(A \cap E_i)$$

$$\Rightarrow P(A) = \sum_{i=1}^n P(E_i \cap A)$$

$$= \sum_{i=1}^n P(E_i) \cdot P(A/E_i) \rightarrow \textcircled{iii}$$



$$\therefore P(A/E_i) = \frac{P(E_i \cap A)}{P(E_i)}$$

Theorem of total probability [viz it gives problem of A from $E_1, E_2, \dots, E_n$ as given conditions].

total
prob. of
A.

Now from eqn. (i) & (ii), we get;

$$P(E_i/A) = \frac{P(E_i) \cdot P(A/E_i)}{\sum_{i=1}^n P(E_i) \cdot P(A/E_i)} \quad \text{which}$$

is the reqd expression of Bayes' theorem.

Note: For calculation, suppose three events E_1, E_2 , and E_3 are given.

Then,

$$\textcircled{i} \quad P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{\sum_{i=1}^3 P(E_i) \cdot P(A/E_i)}$$

$$= \frac{P(E_1) \cdot P(A/E_1)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + P(E_3) \cdot P(A/E_3)}$$

$$\textcircled{ii} \quad P(E_2/A) = \frac{P(E_2) \cdot P(A/E_2)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + P(E_3) \cdot P(A/E_3)}$$

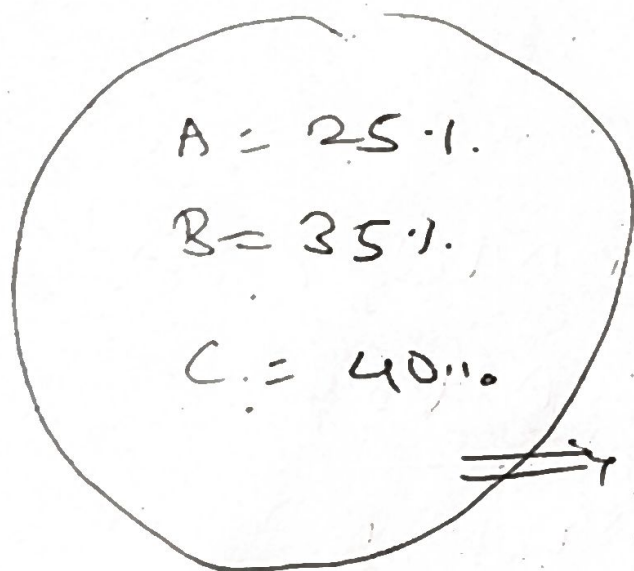
$$\textcircled{iii} \quad P(E_3/A) = \frac{P(E_3) \cdot P(A/E_3)}{\sum_{i=1}^3 P(E_i) \cdot P(A/E_i)}$$

Q) In a bolt factory machine A, B and C manufacture respectively 25%, 35%, and 40% of the total o/p. Out of their o/p respectively 5%, 4% and 2% are defectives. A bolt is selected at random from the total o/p and found to be defective. What is the probability that it is manufactured by;

(i) Machine A

(ii) Machine B

(iii) Machine C



$\Rightarrow$ A bolt is selected

Total o/p = 100%.

Define the events:

E_1 = Bolt manufactured by machine A

E_2 = Bolt from machine B

E_3 = Bolt from machine C

Then, $P(E_1) = 0.25$

$P(E_2) = 0.35$

$P(E_3) = 0.40$

Ans, now. after bolt selection +

bolt is defective.

Additional information.

Let, A - Bolt is defective

Then, $P(A|E_1) = 0.05$

Probability
that Bolt is
defective from
machine A.

$$P(A|E_2) = 0.04$$

$$\& P(A|E_3) = 0.02$$

Probability that the Bolt is defective;

$$P(A) = \sum_{i=1}^3 P(E_i) P(A|E_i)$$

$$= P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) + P(E_3) \cdot P(A|E_3)$$

Then, ~~$P(E_1|A) = P(E_1) \cdot P(A|E_1)$~~ Prob. that

defective bolt is manufactured by

$$P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(A)}$$

prob. that bolt is manufactured by machine A given that it is defective.

$$ii) P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(A)}$$

$$iii) P(E_3|A) = \frac{P(E_3) \cdot P(A|E_3)}{P(A)}$$

es;

1) A manufacturing company produces steel pipes in three plants with daily production volume of 500, 1000, and 2000 units respectively. According to past experience the fraction of defective pipes produced from the three plants are 0.005, 0.008, and 0.01. If a pipe is selected from the day's total production and found to be defective. From which plant the defective pipe is expected to have been produced?

$$\begin{aligned} A &= 500 \\ B &= 1000 \\ C &= 2000 \end{aligned}$$

$$\begin{aligned} \text{Daily total product} &\Rightarrow 500 + 1000 + 2000 \\ &= 3500 \end{aligned}$$

Define the events:

E_1 = steel pipe is produced by plant A

E_2 = " " " " " " B

E_3 = " " " " " " C

Then,

$$P(E_1) = \frac{2000}{3500} = \frac{4}{7}$$

$$P(E_2) = \frac{1000}{3500} = \frac{2}{7}$$

$$P(E_3) = \frac{500}{3500} = \frac{1}{7}$$

Additional info.

A = pipe is defective

Then, $P(A/E_1) = 0.005$

$$P(A/E_2) = 0.008$$

$$P(A/E_3) = 0.01$$

Now, prob. that pipe is defective;

$$P(A) = \sum_{i=1}^3 P(E_i) P(A/E_i)$$

$$= P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + P(E_3) \cdot P(A/E_3)$$

$$= 8.71 \times 10^{-3}$$

Now

i) probability that defective piece is manufactured by plant A is;

$$P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{P(A)}$$

$$= 0.08$$

$$ii) P(E_2/A) = \frac{P(E_2) \cdot P(A/E_2)}{P(A)}$$

$$= 0.26$$

$$iii) P(E_3/A) = \frac{P(E_3) \cdot P(A/E_3)}{P(A)}$$

$$= 0.65$$

Since, $P(E_3/A)$ i.e., probability of defective piece being manufactured by plant C is greatest, so, it's likely to be produced by plant C.

8) The chances of 3 engineers A, B, and C becoming a chief engineer in a certain hydropower company are 4:2:3 respectively. Probability that a new transmission line is introduced if A, B, and C becomes chief engineer are respectively 0.3, 0.5, and 0.8. If a new transmission line has been introduced, what is the probability that;

- (i) A is appointed as a chief engineer
- (ii) B " " " " "
- (iii) C " " " " "

Solⁿ
Defn

Events:

$E_1 =$ ~~chief engineer~~ A is appointed as chief engineer.

$E_2 =$ B is appointed as chief engineer

$E_3 =$ C is appointed as chief engineer.

Now, $P(E_1) = \frac{4}{9}$ { $\because$ Total = $4+2+3 = 9$ }

$P(E_2) = \frac{2}{9}$

$P(E_3) = \frac{3}{9}$

Additional info:

A = a new transmission line is introduced

Then,

$$P(A|E_1) = 0.3$$

$$P(A|E_2) = 0.5$$

$$P(A|E_3) = 0.8$$

Now probability that new transmission line will be introduced;

$$P(A) = \sum_{i=1}^3 P(E_i) P(A|E_i)$$

$$= P(E_1) \cdot P(A|E_1) +$$

$$P(E_2) \cdot P(A|E_2) +$$

$$P(E_3) \cdot P(A|E_3)$$

Now,

(i) A is appointed as chief engineer's probability:

$$P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(A)}$$

(ii) B is appointed as chief engineer's prob,

$$P(E_2|A) = \frac{P(E_2) \cdot P(A|E_2)}{P(A)}$$

(iii) prob. of ~~an~~ ~~appointment~~ being appointed as chief engineer;

$$P(E_3/A) = \frac{P(E_3) \cdot P(A/E_3)}{P(A)} =$$

Q) In a class of 75 students, 15 were considered to be very intelligent, 45 as medium and rest were below the average. The probability that a very intelligent student failing the exam is 0.005, the medium students failing has probability 0.05 and corresponding probability for below average is 0.15. If a student passed the examination what is the probability that he is below average?

Sol.

Define events;

E_1 = a very intelligent ~~student~~ ^{students}

E_2 = ~~best~~ student is medium

E_3 = student is below avg.

So, $P(E_1) = \frac{15}{75}$

$$P(E_2) = \frac{45}{75}$$

$$P(E_3) = \frac{75 - (45 + 15) - 0.05}{75} = \frac{15}{75}$$

Additional Info:

A = student fails

Now,

$$P(A/E_1) = 0.005$$

$$P(A/E_2) = 0.05$$

$$P(A/E_3) = 0.25$$

So,

$$P(A) = \sum_{i=1}^3 P(E_i) \cdot P(A/E_i)$$

$$= P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + P(E_3) \cdot P(A/E_3)$$

$\bar{A}$ = failing the exam

A = passing the exam

↳ this is what is asked to make it A.

Then,

$$P(\bar{A}/E_1) = 0.005 ; P(A/E_1) = 1 - 0.005 = 0.995$$

$$P(\bar{A}/E_2) = 0.05 ; P(A/E_2) = 1 - 0.05 = 0.95$$

$$P(\bar{A}/E_3) = 0.25 ; P(A/E_3) = 1 - 0.25 = 0.75$$

$$\therefore P(A) = \sum_{i=1}^3 P(E_i) \cdot P(A/E_i) = P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + P(E_3) \cdot P(A/E_3)$$

$$S, \quad P(E_3/A) =$$

$$P(E_3) \cdot P(A/E_3)$$

$$P(A)$$

~~≠~~

Q) A given lot of IC chips contains 2% defective chips. Each chip is tested before delivery. The tester itself is not totally reliable. Probability of tester says the chip is good when it is really good is 0.95 and the chip is defective when it is actually defective is 0.94. If a tested device is indicated to be defective, what is the prob. that it is actually defective?

Soln

Define events:

E_1 = Chip is ~~defective~~ actually defective
 E_2 = Chip is defective

Then, $P(E_2) = 0.02 \text{ } \{2\% \}$
 $P(E_1) = 0.98$

Additional info.

$\bar{A}$ = Tester says chip is good
 A = Tester says chip is defective.

$$\Rightarrow P(\bar{A}/E_1) = 0.95$$

$$P(A/E_1) = 0.05 \quad \{1 - 0.95\}$$

$$\Rightarrow P(A/E_2) = 0.94$$

$$\therefore P(\bar{A}/E_2) = 0.06 \quad \{1 - 0.94\}$$

Now,

$$P(A) = \sum_{i=1}^2 P(E_i) \cdot P(A/E_i)$$

$$= P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)$$

Now,

$$P(E_2/A) = \frac{P(E_2) \cdot P(A/E_2)}{P(A)}$$

① The contents of urns I, II, & III are as follows;

- 1 white, 2 Black and 3 red balls;
- 2 white, 1 Black and 1 red balls; and
- 4 white, 5 Black and 3 red balls.

One urn is chosen at random and two balls are ~~chosen~~ drawn. They are found to be white and red. What is the probability that they came from;

(i) Urn I ?

(ii) Urn II ?

(iii) Urn III ?

Urn I	Urn II	Urn III
1W, 2B, 3R	2W, 1B, 1R	4W, 5B, 3R

the urn is chosen

Define the events:

E_1 = Urn I is selected

E_2 = Urn II is chosen

E_3 = Urn III is chosen

Then, $P(E_1) = \frac{1}{3}$

$P(E_2) = \frac{1}{3}$,

$P(E_3) = \frac{1}{3}$.

and now two balls are drawn

↓
Additional info.

Let $A = \text{selecting 1 W and 1 R}$

Then, $P(A/E_1) =$

Total exhaustive cases = $\binom{6}{2} = 15$

& favourable = 1 R & 1 W
 $= {}^3C_1 \times {}^1C_1$

$$\therefore P(A/E_1) = \frac{{}^1C_1 \times {}^3C_1}{\binom{6}{2}}$$

$$\& P(A/E_2) = \frac{{}^2C_1 \times {}^1C_1}{\binom{4}{2}}$$

$$P(A/E_3) = \frac{{}^4C_1 \times {}^3C_1}{\binom{12}{2}}$$

$$P(A) = \sum_{i=1}^3 P(E_i) \cdot P(A/E_i)$$

So,

$$P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{P(A)}$$

✓

Random Variable

eg: Tossing a coin twice $\rightarrow$ Random experiment
Sample space, $S = \{HH, HT, TH, TT\}$

Expressing
in numeric
values

we need a rule for that
 $\Downarrow$
is called. function

eg: $HH \Rightarrow$ rule $\rightarrow$ rule is random variable.
 $\Downarrow$
No. of heads
 $\Downarrow$
then 2.

$HT \rightarrow 1$
 $TH \rightarrow 1$
 $TT \rightarrow 0$

Random Variable (R.V.) is a real valued function
which is associated with each outcomes of
a random experiment.

In other words, R.V. is a function (rule)
which is used to assign a numeric value
for each outcome of a random experiment.

R.V. is denoted by Capital letters $X, Y, Z, \dots$
etc. and the value taken by it is
denoted by small letters $x, y, z, \dots$ etc.

eg,

Tossing a coin twice

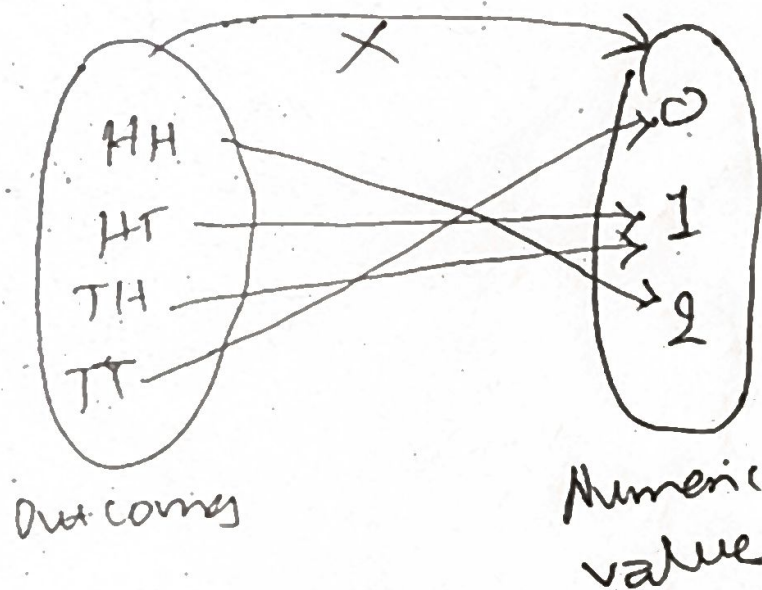
$$S = \{HH, HT, TH, TT\}$$

Consider a random variable, X which

~~$X =$ No. of~~
represents the no. of heads.

i.e., $X =$ No. of heads

Then,



Based on

Numeric value.

Sample space,

$$S = \{0, 1, 2\}$$

i.e.; X can

take the values

either 0 or 1 or

2.

Note:

(i) R.V. is simply the outcomes of
a random experiment.

(ii) R.V. can take any values defined on
sample space of a random experiment.

Types of R.V.

(i) Discrete R.V.

(ii) Continuous R.V.

→ Discrete R.V.

A R.V. is said to be a discrete R.V. if it takes only integers i.e., counting values.

Eg: (i) No. of students enrolled in cosmos per year.

(ii) No. of accidents per day in a busy intersection.

(iii) No. of goals scored in a game.

(iv) No. of printing mistakes in a page.

(v) No. of defective o/p in a given lot.

→ Continuous R.V.

A random variable is said to be continuous R.V. if it takes any values within a certain range or interval.

Eg: (i) Marks obtained by students

(ii) Speed

(iii) Price of a car

(iv) Height of individuals

(v) Temperature

Properties of R.V.

(i) If X is a R.V., then aX , $aX + b$ are also R.V.s where a and b are constants.

(ii) If X and Y are two R.V.s, then $X+Y$, $X-Y$, $+Y$, $+1/Y$ are also R.V.s.

(iii) If X is a R.V. then X^2 , X^3 , ..., X^{-1} , X^{-2} , ... etc. are also R.V.s.

Probability Distribution of R.V.:

The list of probabilities of all possible values of a random variable is called probability distribution.

Ex: Tossing a coin thrice.

$$S = \{HHH, HHT, HTH, THH, TTH, TTT, THT, HTT\}$$

Consider a random variable, X which represents the no. of heads.
i.e., $X = \text{No. of heads}$.

Then, clearly X can take the values;
 $0, 1, 2$, and 3 .

$$\therefore S = \{0, 1, 2, 3\}$$

Now,

$$P(X=0) = \frac{1}{8} = P(\text{No head})$$

$$P(X=1) =$$

$$P(X=1) = \frac{3}{8}$$

$$P(X=2) = \frac{3}{8}$$

$$P(X=3) = \frac{1}{8}$$

$\therefore$ The prob. distribution of X is given as:

$X = x$	$P(X=x)$
0	$\frac{1}{8}$
1	$\frac{3}{8}$
2	$\frac{3}{8}$
3	$\frac{1}{8}$

$\left. \begin{array}{l} \text{Prob. function} \\ \text{Prob. distribution} \end{array} \right\}$

Prob. distribution.

Note:

Probability distribution is the Probability function which is used to calculate the probabilities for all possible values of a random variable.

Type of prob. distribution

- ① Discrete probability distribution
- ② Continuous probability distribution

→ Discrete Prob Distribution

→ Probability distribution of discrete R.V.

→ Probability dist Function of discrete R.V.

→ Probability mass function (pmf)

Let x be a discrete random variable with probability function $P(X=x)$ or $p(x)$; $\forall x$. Then, the probability function $P(X=x)$ is said to be a probability mass function (pmf) which satisfies the following conditions:

① $P(X=x) \geq 0 \Rightarrow$ Non-negative

② $\sum_{-\infty}^{\infty} P(X=x) = 1 \Rightarrow$ Total prob. = 1

eg: A discrete random variable x has the following prob. mass function.

$X=x$	$P(X=x)$
-2	0.1
-1	k
0	0.2
1	2k
2	0.3
3	k

Soln Find the value of k.

We know, the total prob. of prob. distribution is 1.

$$\text{i.e., } \sum_{x=-2}^3 P(X=x) = 1$$

$$\Rightarrow P(X=-2) + P(X=-1) + P(X=0) + P(X=1) + P(X=2) + P(X=3) = 1$$

Probability Distribution Function of Discrete R.V.

$\Rightarrow$ Distribution Function of discrete R.V.
 $\Rightarrow$ Cumulative Distribution function
 (CDF) $\Rightarrow$ Distribution of cumulative probabilities

Let X be a discrete R.V. with probability mass function $P(X=x)$ for all x .

Then, the distribution function of X is denoted by $F(x)$ and is given as:

$$F(x) = P(X \leq x) \\ = \sum_{-\infty}^x P(X=x)$$

eg: in above qn. as $x = 0.1$;

$$F(-2) = P(X \leq -2) = P(-2) = 0.1$$

$$F(-1) = P(X \leq -1) = P(-1) + P(-2)$$

= . . .

$X=x$	$P(X=x)$	$F(x)$ Distribution func.
-2	0.1	0.1 $\Rightarrow F(-2)$
-1	0.1	0.2 $\Rightarrow F(-1)$
0	0.2	0.4 $\Rightarrow F(0)$
1	0.2	0.6
2	0.3	0.9
3	0.1	1

Properties of $F(x)$

(i) $0 \leq F(x) \leq 1$

(ii) $\lim_{x \rightarrow -\infty} F(x) = F(-\infty) = 0$

$\hookrightarrow F(-\infty) = P(X \leq -\infty) = 0$

(iii) $\lim_{x \rightarrow \infty} F(x) = F(\infty) = 1$

$\hookrightarrow F(\infty) = P(X \leq \infty) = \sum_{-\infty}^{\infty} P(X=x) = 1$

(iv) $P(a < X \leq b) = F(b) - F(a)$

(v) if $x > y$ then, $F(x) > F(y)$

Note: For Discrete R.V.,

$P(a < X < b) \neq P(a \leq X \leq b)$

Eg: A discrete R.V. X has the following pmf:

$X=x$	$P(X=x)$
-2	0.1
-1	0.1
0	0.2
1	0.2
2	0.3
3	0.1

(i) Find ~~prob~~ $P(X \leq 1)$

(ii) Find $P(X > 0)$

(iii) Find $P(0 < X < 3)$

(iv) Find $F(-1)$ and $F(2)$

(v) Find the distribution function of X .

Soln

(i) $P(X \leq 1) = P(X = -2) + P(X = -1) + P(X = 0) + P(X = 1)$

(ii) $P(X > 0) = P(X = 1) + P(X = 2) + P(X = 3)$

(iii) $P(0 < X < 3) = P(X = 1) + P(X = 2)$

(iv) $F(-1) = P(X \leq -1) = P(X = -2) + P(X = -1)$

& $F(2) = P(X \leq 2) = P(X = -2) + P(X = -1) + P(X = 0) + P(X = 1) + P(X = 2)$

(v) distribution function of X :

$X = x$	$P(X = x)$	$F(x)$
-2	0.1	0.1
-1	0.1	0.2
0	0.2	0.4
1	0.2	0.6
2	0.3	0.9
3	0.1	1